

Student Grade Calculator Documentation

1. Project Overview and Objectives

Overview

The Student Grade Calculator is a Python-based application that calculates a student's grade based on the marks entered by the user. The program validates user input, assigns grades according to predefined rules, and displays encouraging feedback messages.

Objectives

- Accept student name and marks as input.
 - Calculate grades automatically.
 - Validate marks to ensure they are between 0 and 100.
 - Handle invalid inputs using loops and exception handling.
 - Display motivational messages for each grade.
 - Demonstrate the use of Python functions, conditions, and loops.
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2. Setup and Installation Instructions

Requirements

- Python 3.x installed on your computer.

Installation Steps

1. Download or clone the project repository.
2. Open the project folder.
3. Open Terminal or Command Prompt.
4. Run the program using:

`python grade_calculator.py`

5. Enter the student name and marks when prompted.
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3. Code Structure Explanation

File Structure

Student-Grade-Calculator/

- |
- |— README.md
- |— grade_calculator.py
- |— test_cases.txt
- |— screenshots/

Main Components

Input Section

Collects student name and marks.

```
student_name = input("Enter student name: ")
```

Validation Section

Ensures marks are between 0 and 100.

```
while True:
```

Function Section

Calculates grade and returns a message.

```
def calculate_grade(marks):
```

Output Section

Displays final result to the user.

```
print("Grade:", grade)
```

4. Screenshots of Working Application

Include screenshots in the screenshots folder:

Screenshot 1

Valid Input Example

Enter student name: Priya

Enter marks (0-100): 85

RESULT FOR PRIYA

Marks: 85/100

Grade: B

Message: Very Good! Keep it up!

Screenshot 2

Invalid Marks Example

Enter marks (0-100): 120

Marks must be between 0 and 100

Screenshot 3

Invalid Character Input

Enter marks (0-100): abc

Invalid input. Please enter numbers only.

5. Technical Requirements Fulfilled

Requirement	Implementation
if-elif-else Statements	Used for grade determination
Input Validation	Marks checked between 0 and 100
Function	calculate_grade() function created
Encouraging Messages	Added for every grade
While Loop	Repeats until valid input is entered
Exception Handling	try-except used for invalid input
Testing	Multiple test cases executed

```

grade_calculator.py > ...
1  def calculate_grade(marks):
2      if marks >= 90:
3          return "A", "Excellent Work! Keep shining! 🌟"
4      elif marks >= 80:
5          return "B", "Very Good! Keep it up! 👍"
6      elif marks >= 70:
7          return "C", "Good Job! You can do even better! 😊"
8      elif marks >= 60:
9          return "D", "Keep Working Hard! Success is near! 💪"
10     else:
11         return "F", "Don't Give Up! Practice more and try again! 📚"
12
13
14     student_name = input("Enter student name: ")
15
16
17     while True:
18         try:
19             marks = int(input("Enter marks (0-100): "))
20
21             if 0 <= marks <= 100:
22                 break
23             else:
24                 print("❌ Marks must be between 0 and 100. Try again.")
25
26         except ValueError:
27             print("❌ Invalid input. Please enter numbers only.")
28

```

```

25
26     except ValueError:
27         print("❌ Invalid input. Please enter numbers only.")
28
29
30     grade, message = calculate_grade(marks)
31
32
33     print("\n📋 RESULT FOR", student_name.upper())
34     print("Marks:", marks, "/100")
35     print("Grade:", grade)
36     print("Message:", message)
37

```

Output:

```

📋 RESULT FOR PRIYA
Marks: 85 /100
Grade: B
Message: Very Good! Keep it up! 🍀
PS C:\Users\komal\OneDrive\Desktop\python> & C:\Python314\python.exe c:/Users/komal/OneDrive/Desktop/python/grade_calculator.py
Enter student name: Komal
Enter marks (0-100): 95

📋 RESULT FOR KOMAL
Marks: 95 /100
Grade: A
Message: Excellent Work! Keep shining! 🌟
PS C:\Users\komal\OneDrive\Desktop\python> & C:\Python314\python.exe c:/Users/komal/OneDrive/Desktop/python/grade_calculator.py
Enter student name: Kiran
Enter marks (0-100): 70

📋 RESULT FOR KIRAN
Marks: 70 /100
Grade: C
Message: Good Job! You can do even better! 😊
PS C:\Users\komal\OneDrive\Desktop\python> & C:\Python314\python.exe c:/Users/komal/OneDrive/Desktop/python/grade_calculator.py
Enter student name: Mahesh
Enter marks (0-100): 60

```

```

📋 RESULT FOR KIRAN
Marks: 70 /100
Grade: C
Message: Good Job! You can do even better! 😊
PS C:\Users\komal\OneDrive\Desktop\python> & C:\Python314\python.exe c:/Users/komal/OneDrive/Desktop/python/grade_calculator.py
Enter student name: Mahesh
Enter marks (0-100): 60

📋 RESULT FOR MAHESH
Marks: 60 /100
Grade: D
Message: Keep Working Hard! Success is near! 🍀
PS C:\Users\komal\OneDrive\Desktop\python> & C:\Python314\python.exe c:/Users/komal/OneDrive/Desktop/python/grade_calculator.py
Enter student name: Jiya
Enter marks (0-100): 55

📋 RESULT FOR JIYA
Marks: 55 /100
Grade: F
Message: Don't Give Up! Practice more and try again! 📚

```